All Descendants (Query 01)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
F40	Q_{desc}	Root	9.595	1.427	1.400	6.724x	6.854x	3.043	1.891	1.905	1.609x	1.597x	2.018	2.038	2.301	0.990x	0.877x
F40	Q_{desc}	Leaf parent H	8.774	1.387	1.376	6.326x	6.376x	2.678	1.423	1.270	1.882x	2.109x	1.343	1.559	1.545	0.861x	0.869x
F40	Q_{desc}	Leaf parent	8.862	1.381	1.386	6.417x	6.394x	2.708	1.874	1.662	1.445x	1.629x	1.306	1.523	1.665	0.858x	0.784x
F1K	Q_{desc}	Root	441.595	2.265	2.276	194.965x	194.022x	6.348	3.089	2.515	2.055x	2.524x	29.528	28.457	41.476	1.038x	0.712x
F1K	Q_{desc}	Leaf parent H	13.348	2.280	2.276	5.854x	5.865x	4.194	1.887	2.235	2.223x	1.877x	1.917	2.816	2.256	0.681x	0.850x
F1K	Q_{desc}	Leaf parent	13.020	1.950	1.828	6.677x	7.123x	3.564	2.043	1.963	1.744x	1.816x	1.776	1.725	1.876	1.030x	0.947x
NT10	Q_{desc}	Root	9.267	1.332	1.334	6.957x	6.947x	2.789	1.995	1.693	1.398x	1.647x	2.185	2.128	2.275	1.027x	0.960x
NT10	Q_{desc}	Leaf parent H/L	9.261	1.319	1.349	7.021x	6.865x	2.593	1.738	1.513	1.492x	1.714x	1.430	1.718	1.628	0.832x	0.878x
NT100	Q_{desc}	Root	20.072	1.553	1.516	12.925x	13.240x	3.881	1.726	1.750	2.249x	2.218x	11.480	13.349	13.672	0.860x	0.840x
NT100	Q_{desc}	Leaf parent H	9.863	1.458	1.411	6.765x	6.990x	3.068	1.844	1.580	1.664x	1.942x	2.169	2.372	2.321	0.914x	0.935x
NT100	Q_{desc}	Leaf parent	8.816	1.445	1.427	6.101x	6.178x	2.935	1.517	1.670	1.935x	1.757x	1.516	1.602	1.834	0.946x	0.827x
NT1K	Q_{desc}	Root	1,114.485	2.774	2.849	401.761x	391.185x	10.135	1.959	2.050	5.174x	4.944x	57.664	45.967	47.616	1.254x	1.211x
NT1K	Q_{desc}	Leaf parent H	15.525	2.312	2.285	6.715x	6.794x	2.943	1.539	1.730	1.912x	1.701x	1.329	1.429	1.479	0.930x	0.899x
NT1K	Q_{desc}	Leaf parent	12.488	2.318	2.269	5.387x	5.504x	3.812	1.743	1.794	2.187x	2.125x	1.165	1.177	1.291	0.990x	0.902x
NT10K	Q_{desc}	Root	107,135.344	15.585	15.568	6,874.260x	6,881.767x	101.257	10.707	9.167	9.457x	11.046x	416.490	487.959	533.658	0.854x	0.780x
NT10K	Q_{desc}	Leaf parent H	269.009	11.509	9.715	23.374x	27.690x	2.711	4.809	2.506	0.564x	1.082x	1.442	1.415	1.382	1.019x	1.043x
NT10K	Q_{desc}	Leaf parent	245.370	11.571	9.825	21.206x	24.974x	3.007	5.899	2.163	0.510x	1.390x	1.205	1.792	1.859	0.672x	0.648x
NT100K	Q_{desc}	Root	19,456,928.268	147.652	140.565	131,775.582x	138,419.438x	261.234	91.728	77.402	2.848x	3.375x	4,319.690	5,255.665	5,310.444	0.822x	0.813x
NT100K	Q_{desc}	Leaf parent H	2,542.578	138.030	124.276	18.420x	20.459x	3.794	50.802	22.603	0.075x	0.168x	2.125	2.480	3.287	0.857x	0.646x
NT100K	Q_{desc}	Leaf parent	469.864	137.752	114.226	3.411x	4.113x	2.486	51.423	26.885	0.048x	0.092x	1.448	1.602	1.880	0.904x	0.770x
DT10	Q_{desc}	Root	9.480	1.363	1.406	6.955x	6.743x	2.923	1.471	1.740	1.987x	1.680x	1.155	2.138	2.182	0.540x	0.529x
DT10	Q_{desc}	Leaf parent H	9.312	1.366	1.363	6.817x	6.832x	2.750	1.521	1.562	1.808x	1.761x	1.716	1.891	1.938	0.907x	0.885x
DT10	Q_{desc}	Leaf parent	9.034	1.367	1.351	6.609x	6.687x	2.835	1.565	1.637	1.812x	1.732x	1.308	1.345	1.354	0.972x	0.966x
DT100	Q_{desc}	Root	20.411	1.494	1.477	13.662x	13.819x	3.942	1.538	1.765	2.563x	2.233x	11.579	13.444	13.763	0.861x	0.841x
DT100	Q_{desc}	Leaf parent H	9.298	1.423	1.416	6.534x	6.566x	3.046	1.542	1.662	1.975x	1.833x	1.701	1.791	1.799	0.950x	0.946x
DT100	Q_{desc}	Leaf parent	9.302	1.425	1.451	6.528x	6.411x	2.826	1.597	1.638	1.770x	1.725x	1.487	1.580	1.622	0.941x	0.917x
DT1K	Q_{desc}	Root	1,107.897	2.816	2.845	393.429x	389.419x	11.102	1.694	1.798	6.554x	6.175x	39.750	47.304	49.029	0.840x	0.811x
DT1K	Q_{desc}	Leaf parent H	13.732	2.321	2.233	5.916x	6.150x	3.450	1.680	1.602	2.054x	2.154x	0.865	0.907	0.877	0.954x	0.986x
DT1K	Q_{desc}	Leaf parent	10.918	2.304	2.242	4.739x	4.870x	3.726	1.564	1.536	2.382x	2.426x	0.724	0.788	1.045	0.919x	0.693x
DT10K	Q_{desc}	Root	107,336.589	15.930	15.394	6,738.016x	6,972.625x	32.066	7.073	8.840	4.534x	3.627x	417.619	493.133	505.934	0.847x	0.825x
DT10K	Q_{desc}	Leaf parent H	246.817	11.565	9.793	21.342x	25.203x	2.610	4.647	2.653	0.562x	0.984x	1.241	1.403	1.606	0.885x	0.773x
DT10K	Q_{desc}	Leaf parent	223.195	11.503	9.192	19.403x	24.281x	2.580	5.279	2.807	0.489x	0.919x	1.415	1.529	1.175	0.925x	1.204x
DT100K	Q_{desc}	Root	**>6 h**	143.326	141.381	**>150,705.384x**	**>152,778.661x**	134.708	92.600	77.180	1.455x	1.745x	4,296.936	5,152.568	5,355.181	0.834x	0.802x
DT100K	Q_{desc}	Leaf parent H	801.433	143.799	120.790	5.573x	6.635x	3.368	51.367	26.566	0.066x	0.127x	1.346	1.815	1.780	0.742x	0.756x
DT100K	Q_{desc}	Leaf parent	473.370	138.314	120.321	3.422x	3.934x	2.305	51.005	13.374	0.045x	0.172x	1.599	1.609	1.599	0.994x	1.000x

Queries: Q_{desc} = All Descendants; Q_{child} = All Children; Q_{leaf} = All Leaves; $Q_{a\&d}$ = Check if Ancestor.
Parameters: Root = root of largest tree (also deepest when identical); Deep root = root of deepest tree.
Leaf parent = low-degree leaf parent; Leaf parent H = high-degree leaf parent; H/L = identical high/low-degree cases.
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; $S_D = B / D$; $S_P = B / P$. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.
Timeouts: each >T runtime cell gives the user-confirmed limit T for that row and method (unit shown).
Speedup bounds use those limits in the same row/system; > is a lower bound, < an upper bound.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

All Descendants (Query 01)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
WT10	Q_{desc}	Root	9.419	1.391	1.421	6.771x	6.628x	3.040	1.599	1.545	1.901x	1.968x	2.048	2.198	2.245	0.932x	0.912x
WT10	Q_{desc}	Leaf parent H	9.306	1.402	1.331	6.638x	6.992x	3.089	1.636	1.486	1.888x	2.079x	1.693	1.898	1.995	0.892x	0.849x
WT10	Q_{desc}	Leaf parent	9.040	1.362	1.371	6.637x	6.594x	2.835	1.426	1.456	1.988x	1.947x	1.448	1.498	1.500	0.967x	0.965x
WT100	Q_{desc}	Root	25.971	2.194	1.808	11.837x	14.364x	3.474	1.535	1.506	2.263x	2.307x	11.628	13.204	13.506	0.881x	0.861x
WT100	Q_{desc}	Leaf parent H	10.257	1.450	1.443	7.074x	7.108x	2.652	1.686	1.368	1.573x	1.939x	2.540	2.724	2.778	0.932x	0.914x
WT100	Q_{desc}	Leaf parent	11.713	1.749	1.724	6.697x	6.794x	3.020	1.499	1.545	2.015x	1.955x	1.732	2.101	1.831	0.824x	0.946x
WT1K	Q_{desc}	Root	1,152.546	2.632	2.736	437.897x	421.252x	8.740	1.642	2.225	5.323x	3.928x	58.822	45.133	46.768	1.303x	1.258x
WT1K	Q_{desc}	Leaf parent H	20.250	2.298	2.328	8.812x	8.698x	3.286	1.890	1.816	1.739x	1.809x	1.575	2.044	2.638	0.771x	0.597x
WT1K	Q_{desc}	Leaf parent	21.868	2.260	1.738	9.676x	12.582x	3.498	1.597	1.714	2.190x	2.041x	2.180	2.340	2.475	0.932x	0.881x
WT10K	Q_{desc}	Root	112,250.550	8.642	10.905	12,988.955x	10,293.494x	24.882	10.290	8.581	2.418x	2.900x	407.507	486.496	503.709	0.838x	0.809x
WT10K	Q_{desc}	Leaf parent H	332.930	11.569	10.080	28.778x	33.029x	2.426	5.196	3.014	0.467x	0.805x	1.706	2.189	2.788	0.779x	0.612x
WT10K	Q_{desc}	Leaf parent	114.708	5.704	1.900	20.110x	60.373x	2.958	4.574	2.867	0.647x	1.032x	2.413	2.628	2.598	0.918x	0.929x
WT100K	Q_{desc}	Root	23,750,824.047	76.364	74.205	311,021.215x	320,070.400x	122.530	91.594	75.554	1.338x	1.622x	4,312.966	5,124.128	5,269.493	0.842x	0.818x
WT100K	Q_{desc}	Leaf parent H	2,976.585	139.220	126.679	21.380x	23.497x	3.489	50.023	26.492	0.070x	0.132x	2.347	2.641	2.605	0.889x	0.901x
WT100K	Q_{desc}	Leaf parent	575.098	37.432	1.945	15.364x	295.680x	2.223	50.690	26.894	0.044x	0.083x	1.210	1.306	1.312	0.926x	0.922x
SNB/C	Q_{desc}	Root	91,997.459	680.986	1.951	135.094x	47,154.003x	3.569	416.958	391.391	0.009x	0.009x	4.709	4.490	4.710	1.049x	1.000x
SNB/C	Q_{desc}	Deep root	81,661.090	2,631.127	2,120.512	31.037x	38.510x	4.115	393.337	400.441	0.010x	0.010x	4.076	4.413	4.063	0.924x	1.003x
SNB/C	Q_{desc}	Leaf parent H	68,727.342	2,615.814	2,430.716	26.274x	28.275x	3.632	409.972	366.132	0.009x	0.010x	3.495	3.772	4.002	0.927x	0.873x
SNB/C	Q_{desc}	Leaf parent	22,390.973	724.077	2.021	30.923x	11,079.155x	3.436	436.106	392.928	0.008x	0.009x	1.578	1.577	1.590	1.001x	0.992x
SNB/P	Q_{desc}	Root	16,838.769	2.706	2.559	6,222.753x	6,580.215x	6.908	2.499	2.252	2.764x	3.067x	46.343	41.164	42.907	1.126x	1.080x
SNB/P	Q_{desc}	Leaf parent H	15,130.467	2.987	2.940	5,065.439x	5,146.417x	3.682	1.954	1.817	1.884x	2.026x	10.894	12.844	13.839	0.848x	0.787x
SNB/P	Q_{desc}	Leaf parent	14,506.638	1.606	1.858	9,032.776x	7,807.663x	3.830	2.114	1.876	1.812x	2.042x	1.132	1.304	1.533	0.868x	0.738x
SNB/T	Q_{desc}	Root	14,465.288	1.870	1.900	7,735.448x	7,613.309x	3.337	1.704	1.613	1.958x	2.069x	9.695	10.958	11.340	0.885x	0.855x
SNB/T	Q_{desc}	Leaf parent H	14,542.576	1.472	1.458	9,879.467x	9,974.332x	3.072	1.607	1.698	1.912x	1.809x	8.592	8.915	9.099	0.964x	0.944x
SNB/T	Q_{desc}	Leaf parent	14,497.296	1.605	1.684	9,032.583x	8,608.846x	2.943	1.588	1.507	1.853x	1.953x	1.530	1.521	1.535	1.006x	0.997x

Queries: Q_{desc} = All Descendants; Q_{child} = All Children; Q_{leaf} = All Leaves; $Q_{a\&d}$ = Check if Ancestor.
Parameters: Root = root of largest tree (also deepest when identical); Deep root = root of deepest tree.
Leaf parent = low-degree leaf parent; Leaf parent H = high-degree leaf parent; H/L = identical high/low-degree cases.
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; S_D = *B* / *D*; S_P = *B* / *P*. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

All Children (Query 02)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
F40	Q_{child}	High degree	1.999	1.602	1.494	1.248x	1.338x	1.661	1.822	1.796	0.912x	0.925x	1.664	1.585	1.603	1.050x	1.038x
F40	Q_{child}	Low degree	1.902	1.600	1.493	1.189x	1.274x	1.989	1.668	1.918	1.192x	1.037x	1.207	1.084	1.073	1.113x	1.125x
F1K	Q_{child}	High degree	2.399	2.476	2.315	0.969x	1.036x	1.599	2.545	2.487	0.628x	0.643x	1.924	1.935	1.641	0.994x	1.172x
F1K	Q_{child}	Low degree	2.260	2.371	2.227	0.953x	1.015x	1.490	2.297	1.400	0.649x	1.064x	1.079	1.110	1.129	0.972x	0.956x
NT10	Q_{child}	High degree	1.948	1.584	1.338	1.230x	1.456x	1.651	2.002	1.865	0.825x	0.885x	1.574	1.684	1.669	0.935x	0.943x
NT10	Q_{child}	Low degree	1.912	1.555	1.434	1.230x	1.333x	1.639	1.750	1.281	0.937x	1.279x	1.294	1.109	1.114	1.167x	1.162x
NT100	Q_{child}	High degree	2.003	1.485	1.541	1.349x	1.300x	1.834	1.704	1.693	1.076x	1.083x	2.028	2.079	2.220	0.975x	0.914x
NT100	Q_{child}	Low degree	1.901	1.614	1.497	1.178x	1.270x	1.736	1.738	1.831	0.999x	0.948x	1.259	1.166	1.419	1.080x	0.887x
NT1K	Q_{child}	High degree	2.423	2.407	2.331	1.007x	1.039x	1.713	1.934	1.958	0.886x	0.875x	1.194	1.127	1.137	1.059x	1.050x
NT1K	Q_{child}	Low degree	2.242	2.319	2.213	0.967x	1.013x	1.735	1.912	1.809	0.907x	0.959x	0.708	0.731	0.684	0.969x	1.035x
NT10K	Q_{child}	High degree	6.320	10.517	10.382	0.601x	0.609x	2.155	4.411	2.990	0.489x	0.721x	1.861	1.891	1.854	0.984x	1.004x
NT10K	Q_{child}	Low degree	5.413	10.373	10.257	0.522x	0.528x	2.205	5.085	2.434	0.434x	0.906x	1.097	1.165	1.128	0.942x	0.973x
NT100K	Q_{child}	High degree	57.333	107.338	104.025	0.534x	0.551x	2.214	38.898	27.009	0.057x	0.082x	3.527	3.855	4.112	0.915x	0.858x
NT100K	Q_{child}	Low degree	55.795	105.013	103.651	0.531x	0.538x	1.931	38.570	26.728	0.050x	0.072x	1.313	1.411	1.335	0.931x	0.984x
DT10	Q_{child}	High degree	1.993	1.575	1.481	1.265x	1.346x	1.818	1.376	1.459	1.321x	1.246x	1.491	1.667	1.564	0.894x	0.953x
DT10	Q_{child}	Low degree	1.917	1.552	1.455	1.235x	1.318x	1.578	1.759	1.718	0.897x	0.919x	1.083	1.113	1.116	0.973x	0.970x
DT100	Q_{child}	High degree	2.025	1.642	1.570	1.233x	1.290x	1.354	1.363	1.657	0.993x	0.817x	1.684	1.577	1.517	1.068x	1.110x
DT100	Q_{child}	Low degree	1.950	1.623	1.510	1.201x	1.291x	1.891	1.591	1.790	1.189x	1.056x	1.023	1.089	1.091	0.939x	0.938x
DT1K	Q_{child}	High degree	2.430	2.384	2.284	1.019x	1.064x	1.132	1.771	1.884	0.639x	0.601x	1.510	1.547	1.436	0.976x	1.052x
DT1K	Q_{child}	Low degree	2.253	2.390	2.251	0.943x	1.001x	1.842	1.788	1.811	1.030x	1.017x	0.951	1.020	1.038	0.932x	0.916x
DT10K	Q_{child}	High degree	6.292	10.510	10.320	0.599x	0.610x	1.669	4.870	2.831	0.343x	0.590x	1.483	1.458	1.452	1.017x	1.021x
DT10K	Q_{child}	Low degree	5.367	10.357	10.162	0.518x	0.528x	1.750	4.315	2.612	0.406x	0.670x	1.013	0.987	0.985	1.026x	1.028x
DT100K	Q_{child}	High degree	65.479	105.034	99.786	0.623x	0.656x	1.709	39.166	20.299	0.044x	0.084x	1.786	1.739	1.681	1.027x	1.062x
DT100K	Q_{child}	Low degree	56.285	107.146	103.856	0.525x	0.542x	1.794	38.937	20.222	0.046x	0.089x	1.189	1.316	1.180	0.903x	1.008x

Queries: Q_{desc} = All Descendants; Q_{child} = All Children; Q_{leaf} = All Leaves; $Q_{a\&d}$ = Check if Ancestor.
Parameters: High degree = highest-degree node; Low degree = lowest-degree node.
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; S_D = *B* / *D*; S_P = *B* / *P*. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

All Children (Query 02)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
WT10	Q_{child}	High degree	1.985	1.581	1.462	1.256x	1.358x	2.161	1.771	1.713	1.220x	1.262x	1.676	1.722	1.696	0.973x	0.988x
WT10	Q_{child}	Low degree	1.844	1.560	1.380	1.182x	1.336x	1.666	1.640	1.269	1.016x	1.313x	1.120	1.134	1.183	0.988x	0.947x
WT100	Q_{child}	High degree	2.041	1.656	1.565	1.232x	1.304x	1.572	1.715	1.803	0.917x	0.872x	1.965	2.149	1.829	0.914x	1.074x
WT100	Q_{child}	Low degree	1.953	1.651	1.514	1.183x	1.290x	1.602	1.425	1.487	1.124x	1.077x	1.148	1.165	1.194	0.985x	0.961x
WT1K	Q_{child}	High degree	2.454	2.403	2.401	1.021x	1.022x	1.590	1.795	1.848	0.886x	0.860x	2.334	2.454	2.365	0.951x	0.987x
WT1K	Q_{child}	Low degree	2.294	2.290	2.323	1.002x	0.988x	1.565	1.681	1.568	0.931x	0.998x	1.150	1.213	1.249	0.948x	0.921x
WT10K	Q_{child}	High degree	6.300	10.580	10.437	0.595x	0.604x	1.463	5.010	2.899	0.292x	0.505x	2.606	2.948	2.701	0.884x	0.965x
WT10K	Q_{child}	Low degree	5.365	10.315	10.283	0.520x	0.522x	1.493	4.200	2.586	0.355x	0.577x	1.199	1.229	1.220	0.976x	0.983x
WT100K	Q_{child}	High degree	63.041	101.676	99.621	0.620x	0.633x	1.554	38.942	26.642	0.040x	0.058x	1.529	1.750	1.819	0.874x	0.841x
WT100K	Q_{child}	Low degree	54.175	104.853	99.277	0.517x	0.546x	2.033	38.635	22.414	0.053x	0.091x	0.739	0.678	0.693	1.090x	1.066x
SNB/C	Q_{child}	High degree	1,427.943	3,579.392	3,494.630	0.399x	0.409x	2.849	372.764	387.538	0.008x	0.007x	3.379	3.764	3.816	0.898x	0.885x
SNB/C	Q_{child}	Low degree	1,169.302	3,542.637	3,203.171	0.330x	0.365x	1.796	362.538	356.690	0.005x	0.005x	1.220	1.321	0.990	0.924x	1.232x
SNB/P	Q_{child}	High degree	3.547	3.334	3.073	1.064x	1.154x	2.366	2.269	2.085	1.043x	1.135x	24.419	25.572	26.332	0.955x	0.927x
SNB/P	Q_{child}	Low degree	2.402	2.930	2.543	0.820x	0.945x	1.505	2.421	2.387	0.622x	0.630x	1.216	1.333	1.324	0.912x	0.918x
SNB/T	Q_{child}	High degree	2.149	1.685	1.561	1.275x	1.377x	1.677	1.641	1.784	1.022x	0.940x	3.720	4.177	4.132	0.891x	0.900x
SNB/T	Q_{child}	Low degree	1.941	1.634	1.537	1.188x	1.263x	1.662	1.479	1.651	1.124x	1.007x	1.198	1.061	1.278	1.129x	0.937x

Queries: Q_{desc} = All Descendants; Q_{child} = All Children; Q_{leaf} = All Leaves; $Q_{a\&d}$ = Check if Ancestor.
Parameters: High degree = highest-degree node; Low degree = lowest-degree node.
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; S_D = *B* / *D*; S_P = *B* / *P*. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

All Leaves (Query 05)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
F40	Q_{leaf}	Root	9.607	1.569	2.058	6.123x	4.668x	5.335	3.108	1.654	1.717x	3.226x	1.940	1.857	1.713	1.045x	1.133x
F40	Q_{leaf}	Leaf parent H	9.851	1.510	2.047	6.524x	4.812x	4.607	2.655	2.018	1.735x	2.283x	1.372	1.455	1.458	0.943x	0.941x
F40	Q_{leaf}	Leaf parent	9.593	1.473	2.030	6.513x	4.726x	4.201	3.196	1.932	1.314x	2.174x	1.407	1.466	1.665	0.960x	0.845x
F1K	Q_{leaf}	Root	305.917	18.979	2.393	16.119x	127.838x	7.971	2.957	2.208	2.696x	3.610x	11.306	26.387	26.345	0.428x	0.429x
F1K	Q_{leaf}	Leaf parent H	96.178	2.388	2.255	40.276x	42.651x	5.583	3.473	1.942	1.608x	2.875x	2.329	2.708	2.124	0.860x	1.097x
F1K	Q_{leaf}	Leaf parent	94.975	2.378	2.274	39.939x	41.766x	6.353	2.645	1.766	2.402x	3.597x	1.857	1.928	1.662	0.963x	1.117x
NT10	Q_{leaf}	Root	9.728	1.465	1.413	6.640x	6.885x	3.784	2.768	1.923	1.367x	1.968x	1.786	1.857	1.829	0.962x	0.976x
NT10	Q_{leaf}	Leaf parent H/L	9.016	1.426	1.421	6.323x	6.345x	4.602	2.943	1.945	1.564x	2.366x	1.621	1.745	1.521	0.929x	1.066x
NT100	Q_{leaf}	Root	17.951	2.820	6.264	6.366x	2.866x	6.085	3.096	1.593	1.965x	3.820x	9.444	11.295	9.825	0.836x	0.961x
NT100	Q_{leaf}	Leaf parent H	11.243	1.516	6.097	7.416x	1.844x	5.024	2.904	1.813	1.730x	2.771x	2.194	2.422	2.080	0.906x	1.055x
NT100	Q_{leaf}	Leaf parent	10.885	1.520	6.107	7.161x	1.782x	5.360	2.908	1.609	1.843x	3.331x	1.775	1.779	1.562	0.998x	1.136x
NT1K	Q_{leaf}	Root	946.016	126.084	2.776	7.503x	340.784x	12.502	3.210	2.235	3.895x	5.594x	30.556	34.610	35.149	0.883x	0.869x
NT1K	Q_{leaf}	Leaf parent H	161.370	2.279	2.305	70.807x	70.009x	4.504	3.608	2.002	1.248x	2.250x	1.424	1.661	1.481	0.857x	0.962x
NT1K	Q_{leaf}	Leaf parent	158.675	2.454	2.292	64.660x	69.230x	5.170	3.338	1.697	1.549x	3.047x	1.214	1.184	1.111	1.025x	1.093x
NT10K	Q_{leaf}	Root	92,769.580	12,618.377	14.464	7.352x	6,413.826x	48.544	20.457	7.607	2.373x	6.381x	294.744	338.517	349.448	0.871x	0.843x
NT10K	Q_{leaf}	Leaf parent H	11,983.107	11.646	10.806	1,028.946x	1,108.931x	5.716	17.632	2.433	0.324x	2.349x	1.400	1.857	2.128	0.754x	0.658x
NT10K	Q_{leaf}	Leaf parent	11,977.124	11.676	10.788	1,025.790x	1,110.227x	5.113	16.168	2.298	0.316x	2.225x	1.778	1.939	1.855	0.917x	0.958x
NT100K	Q_{leaf}	Root	19,008,309.331	1,153,555.671	113.815	16.478x	167,010.581x	108.652	203.334	74.057	0.534x	1.467x	4,085.551	4,919.862	5,057.594	0.830x	0.808x
NT100K	Q_{leaf}	Leaf parent H	1,230,113.188	339.497	111.961	3,623.340x	10,986.979x	5.698	156.118	21.053	0.036x	0.271x	2.404	2.489	2.386	0.966x	1.008x
NT100K	Q_{leaf}	Leaf parent	1,227,753.730	338.846	112.308	3,623.338x	10,932.024x	4.645	167.285	20.249	0.028x	0.229x	1.285	1.230	1.048	1.045x	1.226x
DT10	Q_{leaf}	Root	9.363	1.501	1.434	6.238x	6.529x	3.879	2.664	1.682	1.456x	2.306x	1.781	1.828	1.990	0.974x	0.895x
DT10	Q_{leaf}	Leaf parent H	9.606	1.414	1.435	6.793x	6.694x	3.564	2.629	1.593	1.356x	2.237x	1.969	1.937	1.899	1.017x	1.037x
DT10	Q_{leaf}	Leaf parent	9.682	1.306	1.470	7.413x	6.586x	3.911	2.627	1.681	1.489x	2.327x	1.533	1.608	1.488	0.953x	1.030x
DT100	Q_{leaf}	Root	16.156	2.819	5.374	5.731x	3.006x	4.503	2.781	1.816	1.619x	2.480x	6.757	8.957	8.180	0.754x	0.826x
DT100	Q_{leaf}	Leaf parent H	11.149	1.535	5.212	7.263x	2.139x	4.094	2.546	1.668	1.608x	2.454x	1.836	1.945	1.831	0.944x	1.003x
DT100	Q_{leaf}	Leaf parent	10.755	1.520	5.260	7.076x	2.045x	4.082	2.888	1.790	1.413x	2.280x	1.577	1.639	1.562	0.962x	1.010x
DT1K	Q_{leaf}	Root	645.369	119.791	2.635	5.387x	244.922x	13.087	2.798	2.011	4.677x	6.508x	37.618	33.753	30.039	1.115x	1.252x
DT1K	Q_{leaf}	Leaf parent H	97.026	2.418	2.263	40.127x	42.875x	4.762	3.210	2.012	1.483x	2.367x	1.336	1.856	1.859	0.720x	0.719x
DT1K	Q_{leaf}	Leaf parent	96.292	2.361	2.310	40.784x	41.685x	5.142	2.853	1.820	1.802x	2.825x	1.646	1.659	1.611	0.992x	1.022x
DT10K	Q_{leaf}	Root	59,616.817	10,885.079	13.112	5.477x	4,546.737x	30.351	10.582	5.294	2.868x	5.733x	197.457	226.684	229.548	0.871x	0.860x
DT10K	Q_{leaf}	Leaf parent H	6,430.020	11.732	10.546	548.075x	609.712x	3.797	13.187	1.815	0.288x	2.092x	1.275	1.572	1.815	0.811x	0.702x
DT10K	Q_{leaf}	Leaf parent	6,417.641	11.661	10.513	550.351x	610.448x	5.169	17.547	1.920	0.295x	2.692x	1.617	1.611	1.567	1.004x	1.032x
DT100K	Q_{leaf}	Root	11,677,527.224	896,628.634	97.395	13.024x	119,898.632x	135.914	156.547	42.143	0.868x	3.225x	2,118.718	2,564.282	2,614.465	0.826x	0.810x
DT100K	Q_{leaf}	Leaf parent H	617,168.018	344.411	110.806	1,791.952x	5,569.807x	5.926	130.102	11.555	0.046x	0.513x	1.260	1.439	1.280	0.876x	0.984x
DT100K	Q_{leaf}	Leaf parent	617,091.973	341.871	108.481	1,805.043x	5,688.480x	4.480	120.946	12.359	0.037x	0.362x	1.110	1.178	1.026	0.942x	1.082x

Queries: Q_{desc} = All Descendants; Q_{child} = All Children; Q_{leaf} = All Leaves; $Q_{a\&d}$ = Check if Ancestor.
Parameters: Root = root of largest tree (also deepest when identical); Deep root = root of deepest tree.
Leaf parent = low-degree leaf parent; Leaf parent H = high-degree leaf parent; H/L = identical high/low-degree cases.
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; $S_D = B / D$; $S_P = B / P$. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

All Leaves (Query 05)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
WT10	Q_{leaf}	Root	9.599	1.494	1.475	6.425x	6.508x	4.078	2.650	1.556	1.539x	2.621x	1.871	1.946	1.867	0.961x	1.002x
WT10	Q_{leaf}	Leaf parent H	9.190	1.421	1.429	6.467x	6.431x	4.316	2.902	1.867	1.487x	2.312x	1.626	1.853	1.807	0.877x	0.900x
WT10	Q_{leaf}	Leaf parent	9.614	1.440	1.430	6.676x	6.723x	3.718	2.548	1.509	1.459x	2.464x	1.506	1.608	1.517	0.937x	0.993x
WT100	Q_{leaf}	Root	20.322	2.879	7.600	7.059x	2.674x	5.044	2.962	1.876	1.703x	2.689x	7.297	11.256	11.986	0.648x	0.609x
WT100	Q_{leaf}	Leaf parent H	11.977	1.572	7.529	7.619x	1.591x	3.702	2.716	1.919	1.363x	1.929x	2.761	3.025	2.952	0.913x	0.935x
WT100	Q_{leaf}	Leaf parent	11.124	1.557	7.340	7.145x	1.516x	3.775	2.676	1.899	1.411x	1.988x	2.062	2.198	2.093	0.938x	0.985x
WT1K	Q_{leaf}	Root	1,134.133	134.894	2.779	8.408x	408.108x	9.438	3.191	1.922	2.958x	4.911x	45.141	40.338	41.903	1.119x	1.077x
WT1K	Q_{leaf}	Leaf parent H	169.639	2.435	2.333	69.667x	72.713x	4.736	3.468	1.944	1.366x	2.436x	1.714	1.820	1.747	0.942x	0.981x
WT1K	Q_{leaf}	Leaf parent	167.408	2.435	2.320	68.751x	72.159x	4.835	3.355	1.619	1.441x	2.986x	1.520	1.617	1.556	0.940x	0.977x
WT10K	Q_{leaf}	Root	110,118.324	13,540.194	15.277	8.133x	7,208.112x	29.754	25.251	8.137	1.178x	3.657x	376.105	400.557	460.990	0.939x	0.816x
WT10K	Q_{leaf}	Leaf parent H	12,540.864	11.762	10.944	1,066.219x	1,145.912x	3.720	21.089	2.488	0.176x	1.495x	2.054	2.630	3.208	0.781x	0.640x
WT10K	Q_{leaf}	Leaf parent	12,518.364	11.643	10.912	1,075.184x	1,147.211x	4.497	21.026	2.534	0.214x	1.775x	2.772	2.658	2.153	1.043x	1.288x
WT100K	Q_{leaf}	Root	18,195,767.103	1,084,120.519	114.491	16.784x	158,927.489x	114.859	197.623	73.464	0.581x	1.563x	3,990.915	4,720.806	4,920.782	0.845x	0.811x
WT100K	Q_{leaf}	Leaf parent H	1,227,203.338	341.271	105.587	3,595.979x	11,622.675x	5.437	160.884	26.350	0.034x	0.206x	2.341	3.010	4.328	0.778x	0.541x
WT100K	Q_{leaf}	Leaf parent	1,224,686.101	337.399	111.792	3,629.786x	10,955.042x	4.340	171.799	20.687	0.025x	0.210x	1.803	2.014	1.921	0.895x	0.939x
SNB/C	Q_{leaf}	Root	**>6 h**	2,686.814	2,345.178	**>8,039.261x**	**>9,210.388x**	6.036	552.892	326.018	0.011x	0.019x	2.048	3.115	2.924	0.657x	0.700x
SNB/C	Q_{leaf}	Deep root	**>6 h**	2,719.554	2,333.129	**>7,942.478x**	**>9,257.953x**	5.859	569.697	293.923	0.010x	0.020x	2.668	2.944	3.104	0.906x	0.860x
SNB/C	Q_{leaf}	Leaf parent H	**>6 h**	2,685.211	2,340.759	**>8,044.060x**	**>9,227.776x**	5.228	592.005	339.899	0.009x	0.015x	3.629	3.919	3.921	0.926x	0.926x
SNB/C	Q_{leaf}	Leaf parent	**>6 h**	2,718.649	2,332.790	**>7,945.122x**	**>9,259.298x**	5.883	584.010	342.576	0.010x	0.017x	1.627	1.665	1.595	0.977x	1.020x
SNB/P	Q_{leaf}	Root	16,988.983	80.895	3.217	210.013x	5,281.002x	9.585	4.264	2.508	2.248x	3.822x	46.143	40.154	41.466	1.149x	1.113x
SNB/P	Q_{leaf}	Leaf parent H	15,466.619	8.773	2.952	1,762.979x	5,239.370x	5.074	3.979	2.100	1.275x	2.416x	11.650	14.298	15.432	0.815x	0.755x
SNB/P	Q_{leaf}	Leaf parent	14,814.599	2.925	2.797	5,064.820x	5,296.603x	5.030	4.218	2.316	1.193x	2.172x	1.237	1.546	1.554	0.800x	0.796x
SNB/T	Q_{leaf}	Root	14,533.593	2.268	4.382	6,408.110x	3,316.657x	4.607	2.476	1.751	1.861x	2.631x	10.191	9.756	9.517	1.045x	1.071x
SNB/T	Q_{leaf}	Leaf parent H	14,546.129	1.991	4.325	7,305.941x	3,363.267x	4.401	2.554	1.679	1.723x	2.621x	8.152	6.093	8.575	1.338x	0.951x
SNB/T	Q_{leaf}	Leaf parent	14,430.295	1.512	4.238	9,543.846x	3,404.978x	3.401	2.422	1.724	1.404x	1.973x	1.656	1.522	1.469	1.088x	1.127x

Queries: Q_{desc} = All Descendants; Q_{child} = All Children; Q_{leaf} = All Leaves; $Q_{a\&d}$ = Check if Ancestor.
Parameters: Root = root of largest tree (also deepest when identical); Deep root = root of deepest tree.
Leaf parent = low-degree leaf parent; Leaf parent H = high-degree leaf parent; H/L = identical high/low-degree cases.
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; $S_D = B / D$; $S_P = B / P$. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.
Timeouts: each >T runtime cell gives the user-confirmed limit T for that row and method (unit shown).
Speedup bounds use those limits in the same row/system; > is a lower bound, < an upper bound.

Speedup color:



gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

Check if Ancestor (Query 11)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
F40	$Q_{a\&d}$	Root–leaf	10.127	1.380	1.289	7.338x	7.856x	22.247	0.904	0.936	24.610x	23.768x	1.476	1.452	1.314	1.017x	1.123x
F40	$Q_{a\&d}$	Deep parent–leaf	10.135	1.099	1.246	9.222x	8.134x	19.873	0.896	0.933	22.180x	21.300x	1.310	1.342	1.504	0.976x	0.871x
F40	$Q_{a\&d}$	Siblings	9.905	1.100	1.086	9.005x	9.121x	20.082	0.974	0.908	20.618x	22.117x	1.538	1.274	1.343	1.207x	1.145x
F40	$Q_{a\&d}$	Distant leaves	9.294	1.387	1.098	6.701x	8.464x	21.392	0.899	0.941	23.795x	22.733x	1.457	1.381	1.407	1.055x	1.036x
F1K	$Q_{a\&d}$	Root–leaf	12.471	1.419	1.477	8.789x	8.443x	19.079	1.005	0.911	18.984x	20.943x	1.730	1.137	1.551	1.522x	1.115x
F1K	$Q_{a\&d}$	Deep parent–leaf	12.246	1.628	1.455	7.522x	8.416x	21.627	0.996	0.920	21.714x	23.508x	1.849	1.172	1.557	1.578x	1.188x
F1K	$Q_{a\&d}$	Siblings	10.838	2.579	2.562	4.202x	4.230x	16.886	1.082	0.878	15.606x	19.232x	1.563	1.257	1.623	1.243x	0.963x
F1K	$Q_{a\&d}$	Distant leaves	10.372	2.523	2.593	4.111x	4.000x	22.305	0.962	1.046	23.186x	21.324x	1.665	1.331	1.630	1.251x	1.021x
NT10	$Q_{a\&d}$	Root–leaf	9.783	1.223	1.074	7.999x	9.109x	18.744	0.945	0.942	19.835x	19.898x	1.555	1.280	1.170	1.215x	1.329x
NT10	$Q_{a\&d}$	Deep parent–leaf	9.847	1.112	1.243	8.855x	7.922x	19.022	0.999	0.951	19.041x	20.002x	1.305	1.172	1.127	1.113x	1.158x
NT10	$Q_{a\&d}$	Siblings	9.314	1.268	0.982	7.345x	9.485x	20.854	1.001	0.976	20.833x	21.367x	1.454	1.430	1.679	1.017x	0.866x
NT10	$Q_{a\&d}$	Distant leaves	9.391	0.998	0.995	9.410x	9.438x	20.381	0.893	0.936	22.823x	21.775x	1.402	1.161	1.238	1.208x	1.132x
NT100	$Q_{a\&d}$	Root–leaf	9.955	1.311	1.304	7.593x	7.634x	20.681	0.950	0.910	21.769x	22.726x	2.025	1.381	0.753	1.466x	2.689x
NT100	$Q_{a\&d}$	Deep parent–leaf	10.085	1.283	1.303	7.860x	7.740x	20.654	0.930	0.981	22.209x	21.054x	1.399	1.296	1.464	1.079x	0.956x
NT100	$Q_{a\&d}$	Siblings	9.307	1.157	1.140	8.044x	8.164x	18.744	0.947	0.926	19.793x	20.242x	1.710	1.349	1.172	1.268x	1.459x
NT100	$Q_{a\&d}$	Distant leaves	9.514	1.151	1.114	8.266x	8.540x	21.837	0.915	0.945	23.866x	23.108x	1.164	0.891	0.879	1.306x	1.324x
NT1K	$Q_{a\&d}$	Root–leaf	10.842	1.464	1.376	7.406x	7.879x	19.521	0.957	0.891	20.398x	21.909x	1.231	0.820	0.713	1.501x	1.727x
NT1K	$Q_{a\&d}$	Deep parent–leaf	10.592	1.396	1.315	7.587x	8.055x	19.318	0.827	0.892	23.359x	21.657x	0.858	1.096	1.093	0.783x	0.785x
NT1K	$Q_{a\&d}$	Siblings	10.784	2.551	2.761	4.227x	3.906x	22.428	0.952	0.865	23.559x	25.928x	1.407	1.093	1.178	1.287x	1.194x
NT1K	$Q_{a\&d}$	Distant leaves	10.589	2.534	2.568	4.179x	4.123x	20.607	0.892	0.950	23.102x	21.692x	1.431	1.060	0.932	1.350x	1.535x
NT10K	$Q_{a\&d}$	Root–leaf	18.172	1.568	1.371	11.589x	13.255x	22.460	0.875	0.925	25.669x	24.281x	1.580	1.271	1.373	1.243x	1.151x
NT10K	$Q_{a\&d}$	Deep parent–leaf	18.112	1.451	1.576	12.482x	11.492x	22.484	0.856	0.871	26.266x	25.814x	1.349	1.132	1.171	1.192x	1.152x
NT10K	$Q_{a\&d}$	Siblings	21.235	16.436	16.269	1.292x	1.305x	21.005	0.890	0.897	23.601x	23.417x	1.364	1.198	1.280	1.139x	1.066x
NT10K	$Q_{a\&d}$	Distant leaves	20.976	16.493	16.307	1.272x	1.286x	24.704	0.836	0.946	29.550x	26.114x	1.567	1.243	1.242	1.261x	1.262x
NT100K	$Q_{a\&d}$	Root–leaf	78.542	1.592	1.628	49.335x	48.244x	19.771	1.108	0.861	17.844x	22.963x	1.144	0.953	0.845	1.200x	1.354x
NT100K	$Q_{a\&d}$	Deep parent–leaf	78.650	1.593	1.593	49.372x	49.372x	22.175	0.947	0.907	23.416x	24.449x	0.930	0.796	0.828	1.168x	1.123x
NT100K	$Q_{a\&d}$	Siblings	125.413	157.166	157.755	0.798x	0.795x	18.425	1.008	0.890	18.279x	20.702x	0.840	0.841	0.814	0.999x	1.032x
NT100K	$Q_{a\&d}$	Distant leaves	125.531	159.396	158.780	0.788x	0.791x	19.703	1.013	0.937	19.450x	21.028x	0.899	0.787	0.849	1.142x	1.059x
DT10	$Q_{a\&d}$	Root–leaf	9.830	1.338	1.328	7.347x	7.402x	17.372	0.887	0.959	19.585x	18.115x	1.441	1.244	1.109	1.158x	1.299x
DT10	$Q_{a\&d}$	Deep parent–leaf	9.698	1.081	1.274	8.971x	7.612x	18.375	0.874	0.997	21.024x	18.430x	1.284	1.153	1.157	1.114x	1.110x
DT10	$Q_{a\&d}$	Siblings	9.365	1.059	1.018	8.843x	9.199x	19.062	0.855	0.971	22.295x	19.631x	1.189	1.187	1.163	1.002x	1.022x
DT10	$Q_{a\&d}$	Distant leaves	9.448	1.025	0.985	9.218x	9.592x	18.044	0.871	0.924	20.716x	19.528x	1.300	1.178	1.199	1.104x	1.084x
DT100	$Q_{a\&d}$	Root–leaf	9.660	1.139	1.329	8.481x	7.269x	20.454	0.839	0.901	24.379x	22.701x	1.483	1.116	1.091	1.329x	1.359x
DT100	$Q_{a\&d}$	Deep parent–leaf	10.179	1.360	1.348	7.485x	7.551x	21.765	0.939	0.932	23.179x	23.353x	1.220	1.122	1.165	1.087x	1.047x
DT100	$Q_{a\&d}$	Siblings	9.524	1.194	1.150	7.977x	8.282x	19.434	1.016	0.887	19.128x	21.910x	1.271	1.174	1.231	1.083x	1.032x
DT100	$Q_{a\&d}$	Distant leaves	9.572	1.175	1.130	8.146x	8.471x	18.781	0.942	0.921	19.937x	20.392x	1.439	1.293	0.671	1.113x	2.145x
DT1K	$Q_{a\&d}$	Root–leaf	10.952	1.439	1.273	7.611x	8.603x	20.319	0.853	0.980	23.821x	20.734x	1.834	1.456	1.187	1.260x	1.545x
DT1K	$Q_{a\&d}$	Deep parent–leaf	10.826	1.248	1.203	8.675x	8.999x	19.947	0.942	0.941	21.175x	21.198x	1.330	1.366	1.191	0.974x	1.117x
DT1K	$Q_{a\&d}$	Siblings	11.192	2.658	2.547	4.211x	4.394x	17.540	0.902	0.916	19.446x	19.148x	1.353	1.263	1.314	1.071x	1.030x
DT1K	$Q_{a\&d}$	Distant leaves	10.726	2.602	2.538	4.122x	4.226x	20.111	0.967	0.939	20.797x	21.417x	1.577	1.336	1.330	1.180x	1.186x
DT10K	$Q_{a\&d}$	Root–leaf	18.656	1.619	1.621	11.523x	11.509x	21.870	0.910	1.044	24.033x	20.948x	1.646	1.217	1.128	1.353x	1.459x
DT10K	$Q_{a\&d}$	Deep parent–leaf	17.961	1.514	1.572	11.863x	11.426x	20.130	0.937	0.908	21.483x	22.170x	1.349	1.099	1.094	1.227x	1.233x
DT10K	$Q_{a\&d}$	Siblings	21.468	16.841	16.706	1.275x	1.285x	19.828	0.866	1.010	22.896x	19.632x	1.398	1.222	1.229	1.144x	1.138x
DT10K	$Q_{a\&d}$	Distant leaves	21.209	16.828	16.731	1.260x	1.268x	20.317	0.976	0.953	20.817x	21.319x	1.592	1.218	1.218	1.307x	1.307x
DT100K	$Q_{a\&d}$	Root–leaf	80.013	1.633	1.589	48.998x	50.354x	19.689	1.041	1.034	18.914x	19.042x	1.130	0.998	0.849	1.132x	1.331x
DT100K	$Q_{a\&d}$	Deep parent–leaf	80.795	1.356	1.338	59.583x	60.385x	21.310	1.024	0.928	20.811x	22.963x	0.852	0.822	1.140	1.036x	0.747x
DT100K	$Q_{a\&d}$	Siblings	130.836	160.346	155.348	0.816x	0.842x	20.041	0.974	0.972	20.576x	20.618x	1.166	1.234	1.264	0.945x	0.922x
DT100K	$Q_{a\&d}$	Distant leaves	130.236	162.187	156.947	0.803x	0.830x	21.146	0.860	0.909	24.588x	23.263x	1.429	1.320	1.303	1.083x	1.097x

Queries: Q_{desc} = All Descendants; Q_{child} = All Children; Q_{leaf} = All Leaves; $Q_{a\&d}$ = Check if Ancestor.
Parameters: Root–leaf = root-farthest-leaf pair; Deep parent–leaf = deepest parent-leaf pair (both true).
Siblings = shallow sibling pair; Distant leaves = leaf pair with greatest distance via their lowest common ancestor (both false).
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; $S_D = B / D$; $S_P = B / P$. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

Check if Ancestor (Query 11)

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
WT10	$Q_{a\&d}$	Root–leaf	9.964	1.337	1.275	7.453x	7.815x	18.427	0.907	0.878	20.316x	20.987x	1.834	1.121	1.121	1.636x	1.636x
WT10	$Q_{a\&d}$	Deep parent–leaf	9.955	1.326	1.312	7.508x	7.588x	18.861	0.848	0.966	22.242x	19.525x	1.265	1.169	1.186	1.082x	1.067x
WT10	$Q_{a\&d}$	Siblings	9.408	1.036	0.979	9.081x	9.610x	18.511	0.870	0.919	21.277x	20.143x	1.360	1.321	1.351	1.030x	1.007x
WT10	$Q_{a\&d}$	Distant leaves	9.701	0.994	0.974	9.760x	9.960x	17.650	0.908	0.918	19.438x	19.227x	1.485	1.322	1.289	1.123x	1.152x
WT100	$Q_{a\&d}$	Root–leaf	11.364	1.706	1.602	6.661x	7.094x	19.110	0.917	0.941	20.840x	20.308x	1.384	0.988	1.103	1.401x	1.255x
WT100	$Q_{a\&d}$	Deep parent–leaf	11.087	1.473	1.465	7.527x	7.568x	18.289	0.899	0.943	20.344x	19.394x	1.225	1.098	1.255	1.116x	0.976x
WT100	$Q_{a\&d}$	Siblings	9.769	1.174	1.270	8.321x	7.692x	17.645	0.918	0.925	19.221x	19.076x	1.289	1.128	1.119	1.143x	1.152x
WT100	$Q_{a\&d}$	Distant leaves	9.785	1.136	1.109	8.614x	8.823x	21.164	0.930	0.975	22.757x	21.707x	1.671	1.372	1.363	1.218x	1.226x
WT1K	$Q_{a\&d}$	Root–leaf	12.688	1.564	1.534	8.113x	8.271x	20.703	0.844	0.891	24.530x	23.236x	1.060	0.805	1.127	1.317x	0.941x
WT1K	$Q_{a\&d}$	Deep parent–leaf	12.054	1.544	1.509	7.807x	7.988x	18.594	0.850	0.885	21.875x	21.010x	1.318	1.283	1.324	1.027x	0.995x
WT1K	$Q_{a\&d}$	Siblings	10.784	2.564	2.518	4.206x	4.283x	20.958	0.881	1.083	23.789x	19.352x	1.443	1.370	1.405	1.053x	1.027x
WT1K	$Q_{a\&d}$	Distant leaves	10.561	2.567	2.521	4.114x	4.189x	22.560	0.936	0.902	24.103x	25.011x	1.578	1.377	1.351	1.146x	1.168x
WT10K	$Q_{a\&d}$	Root–leaf	19.108	1.724	1.774	11.084x	10.771x	21.210	0.860	0.859	24.663x	24.692x	1.386	1.200	0.910	1.155x	1.523x
WT10K	$Q_{a\&d}$	Deep parent–leaf	18.490	1.753	1.732	10.548x	10.676x	21.004	0.959	0.929	21.902x	22.609x	1.230	1.121	1.136	1.097x	1.083x
WT10K	$Q_{a\&d}$	Siblings	20.521	16.425	16.231	1.249x	1.264x	19.295	0.890	0.831	21.680x	23.219x	1.338	1.206	1.204	1.109x	1.111x
WT10K	$Q_{a\&d}$	Distant leaves	20.925	16.604	16.375	1.260x	1.278x	20.818	0.825	0.826	25.234x	25.203x	1.429	1.222	1.246	1.169x	1.147x
WT100K	$Q_{a\&d}$	Root–leaf	79.819	1.898	1.811	42.054x	44.075x	20.214	0.937	0.876	21.573x	23.075x	1.577	1.298	1.294	1.215x	1.219x
WT100K	$Q_{a\&d}$	Deep parent–leaf	81.240	1.833	1.790	44.321x	45.385x	22.522	0.844	0.892	26.685x	25.249x	1.282	1.261	1.329	1.017x	0.965x
WT100K	$Q_{a\&d}$	Siblings	126.201	162.481	157.316	0.777x	0.802x	19.990	0.876	0.905	22.820x	22.088x	1.235	1.292	1.389	0.956x	0.889x
WT100K	$Q_{a\&d}$	Distant leaves	125.904	160.417	159.402	0.785x	0.790x	19.570	0.881	0.930	22.213x	21.043x	1.445	1.427	1.468	1.013x	0.984x
SNB/C	$Q_{a\&d}$	Root–leaf	14,648.588	2.020	1.860	7,251.776x	7,875.585x	22.268	1.035	0.926	21.515x	24.048x	1.557	1.190	1.084	1.308x	1.436x
SNB/C	$Q_{a\&d}$	Deep parent–leaf	14,582.521	2.119	1.814	6,881.794x	8,038.876x	19.387	1.274	1.007	15.217x	19.252x	1.161	1.063	1.126	1.092x	1.031x
SNB/C	$Q_{a\&d}$	Siblings	15,882.238	3,576.593	3,516.908	4.441x	4.516x	18.685	1.057	0.856	17.677x	21.828x	1.167	1.132	1.172	1.031x	0.996x
SNB/C	$Q_{a\&d}$	Distant leaves	15,854.278	3,769.164	3,664.008	4.206x	4.327x	20.317	1.091	0.848	18.622x	23.959x	1.212	1.078	1.116	1.124x	1.086x
SNB/P	$Q_{a\&d}$	Root–leaf	14,525.595	1.120	1.493	12,969.281x	9,729.133x	19.577	0.801	0.892	24.441x	21.947x	1.421	1.116	1.079	1.273x	1.317x
SNB/P	$Q_{a\&d}$	Deep parent–leaf	14,422.871	1.565	1.481	9,215.892x	9,738.603x	17.314	0.882	0.908	19.630x	19.068x	1.267	1.097	1.103	1.155x	1.149x
SNB/P	$Q_{a\&d}$	Siblings	14,476.265	3.228	3.172	4,484.593x	4,563.766x	20.944	0.870	0.917	24.074x	22.840x	1.202	1.115	1.178	1.078x	1.020x
SNB/P	$Q_{a\&d}$	Distant leaves	14,341.406	3.168	3.137	4,526.959x	4,571.695x	17.612	0.878	0.902	20.059x	19.525x	1.296	1.182	1.193	1.096x	1.086x
SNB/T	$Q_{a\&d}$	Root–leaf	14,538.547	1.608	1.687	9,041.385x	8,617.989x	18.943	0.824	0.946	22.989x	20.024x	1.204	1.146	1.200	1.051x	1.003x
SNB/T	$Q_{a\&d}$	Deep parent–leaf	14,437.473	1.439	1.679	10,032.990x	8,598.852x	20.531	0.832	0.876	24.677x	23.437x	1.389	1.109	1.091	1.252x	1.273x
SNB/T	$Q_{a\&d}$	Siblings	14,251.434	1.063	1.019	13,406.805x	13,985.706x	19.291	0.834	0.894	23.131x	21.578x	1.205	1.112	1.205	1.084x	1.000x
SNB/T	$Q_{a\&d}$	Distant leaves	14,278.768	1.007	0.987	14,179.511x	14,466.837x	18.252	0.848	0.825	21.524x	22.124x	1.275	1.469	1.155	0.868x	1.104x

Queries: Q_{desc} = All Descendants; Q_{child} = All Children; Q_{leaf} = All Leaves; $Q_{a\&d}$ = Check if Ancestor.
Parameters: Root–leaf = root-farthest-leaf pair; Deep parent–leaf = deepest parent-leaf pair (both true).
Siblings = shallow sibling pair; Distant leaves = leaf pair with greatest distance via their lowest common ancestor (both false).
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; $S_D = B / D$; $S_P = B / P$. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup

LDBC SNB SF1: IC12, IS2, IS6

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P	*B*	*D*	*P*	S_D	S_P
SNB	IC12	–	188,529.785	170,910.577	173,980.686	1.103x	1.084x	396.524	304.851	403.867	1.301x	0.982x	157.466	119.408	106.318	1.319x	1.481x
SNB	IS2	–	61,383.859	3,842.891	13,060.742	15.973x	4.700x	43.709	242.223	277.390	0.180x	0.158x	4.510	4.535	10.652	0.994x	0.423x
SNB	IS6	–	17,296.910	2,172.181	3,269.589	7.963x	5.290x	28.813	22.864	118.383	1.260x	0.243x	1.897	2.004	1.817	0.947x	1.044x

Queries: IC12 = Interactive Complex 12; IS2 = Interactive Short 2; IS6 = Interactive Short 6.
Methods: *B* = Baseline; *D* = Dewey; *P* = PrePost; S_D = *B* / *D*; S_P = *B* / *P*. Runtimes: median of five runs in ms.
Graphs: F = forest; NT = truebase; DT = ultratall; WT = ultrawide; K = 1,000 nodes.
SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Speedup color: 

gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup